An Explainable Deep Learning with Transfer Learning Approach for Lung Cancer Detection

by

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Thesis (CSE 400) submitted in partial fulfillment of the requirements for the degree of

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Under the supervision of

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DECLARATION

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We declare that this thesis entitled *An Explainable Deep Learning with Transfer Learning Approach for Lung Cancer Detection* is the result of our own work except as cited in the references. The thesis has not been accepted for any degree and is not concurrently submitted in candidature of any other degree.

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Dedicated to
my loving Mother

– Md. Refayet Hossen

To my beloved parents
the source of my inspiration

– Priyo Ghosh

To my parents
for their love and
constant encouragement.

– Saiful Kobir Chowdhury

ABSTRACT

Lung cancer is one of the top reasons people die from cancer around the world. Finding it early is very important to help patients live longer. Lately, deep learning methods have shown a lot of promise in looking at medical images to help with diagnosis. This study presents a deep learning approach to find lung cancer using CT scan pictures. The proposed system uses a transfer learning method based on a model called ResNet18 to sort lung images into three types: normal, benign, and malignant. The model is trained on the IQ-OTH/NCCD dataset after basic steps like resizing images and normalizing them. The dataset is split into training, validation, and testing sets to make sure the evaluation is fair. Test results show that the model works very well, reaching an overall accuracy of 99%. To make the model easier to understand, Grad-CAM is used to highlight the important areas that influence the model's decisions. The results show that the model focuses on meaningful parts of the lungs, which makes its predictions clearer and more useful for doctors. Because it is both highly accurate and easy to interpret, the proposed method could be a helpful tool for doctors diagnosing lung cancer. This work helps in building an effective and understandable automated system for detecting cancer from medical images.

Keywords: Lung cancer detection, Deep learning, ResNet18, Transfer learning, Grad-CAM, Medical image analysis

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1. Introduction

1.1 Overview

Lung cancer is a major contributor to cancer-related deaths globally, making up a large fraction of cancer fatalities each year [1, 2]. Timely and accurate diagnostic methods can be lifesaving, as early detection leads to appropriate and targeted treatment [3]. Various medical imaging technologies, such as computed tomography (CT) scans and chest X-rays, are employed to facilitate timely disease diagnosis. Recent progress in artificial intelligence, particularly in deep learning [4], enables the automation of medical imaging analysis with exceptional accuracy [5].Convolutional neural networks (CNNs) are particularly efficient at identifying complex visual features, which makes them useful for classifying medical images. Despite their effectiveness, many existing deep learning-based models are still considered black boxes [6], meaning they lack clarity and make it difficult to understand how decisions are formulated. Explainable artificial intelligence (XAI) has emerged as a crucial area of research aimed at enhancing the transparency and interpretability of models [7]. In healthcare, this aspect is vital because physicians need to comprehend the reasoning behind the system's decisions before placing their trust in it. One approach that aids in clarifying the outputs of deep neural networks is the Gradient-weighted Class Activation Mapping (Grad-CAM) technique [8], which highlights important areas of the image.

This study develops a deep learning-based algorithm to classify lung images into three categories: normal, benign, or malignant.. At the same time, explainable AI algorithms will be applied to produce interpretable visual explanations of predictions.

1.2 Background

Lung cancer poses a significant public health problem as an uncontrollable cellular growth in the lungs' tissues [9]. There exist two categories of conditions – benign and malignant. The latter is associated with cancer development, whereas benign lung disease involves non-cancerous abnormalities that may also require clinical treatment [10]. Survival chances are increased when a condition is diagnosed early, thus accurate classification of a lung disease is vital for effective diagnosis and treatment [2]. Diagnosis of lung cancer depends on medical imaging since it allows physicians to detect abnormalities in the lung area. Yet, the interpretation of the images may be challenging and subjective. Hence, computer-aided diagnosis (CAD) systems were introduced for automatic image analysis that would help radiologists make decisions based on visual examinations [5].

The recent improvements in the field of deep learning and especially in Convolutional Neural Networks (CNN) led to great success in the automated image classification task [11]. The algorithms are able to classify between three classes – normal, benign, and malignant lung images; however, they lack the interpretability [12]. Explainable Artificial Intelligence provides transparency concerning the prediction results obtained via neural networks. Grad-CAM produces the image that demonstrates the part of the input image that is responsible for making the prediction [8]. It is important for the medical application domain because AI should provide reliable solutions to health specialists [13].

1.3 Problem Domain and Motivation

Lung cancer diagnosis based on medical imaging poses several difficulties owing to variations in the dimensions, shape, texture, and location of lesions [14]. Radiologists apply their knowledge and skills in interpreting the image results, which can be time-consuming and susceptible to bias [15]. Automated models using deep learning technology can help by providing reliable and fast classification [5]. Despite their excellent results, deep learning models generally lack interpretability, thus limiting their implementation in a real-world environment [16]. An interpretability-free model can prove unreliable in crucial fields like medicine because it does not provide an explanation for its decision-making processes. The

lack of transparency leads to a discrepancy between the model's actual performance and applicability. In addition, the differentiation between benign and malignant cases and normal conditions is necessary for accurate diagnosis and treatment planning since misclassification will cause late treatments or unnecessary operations. Therefore, there exists a necessity for a model that accurately classifies lung cancer cases and provides interpretable results.

The motivation behind this research is to develop a diagnostic tool that incorporates explainable artificial intelligence methods. This technology allows users to generate explanations for model decisions graphically.

1.4 Research Questions

This study is guided by the following research questions:

- Is it possible for a deep learning system to correctly detect lung images as normal, benign, or malignant?
- How can the techniques of explainable artificial intelligence help in the interpretation of the model's predictions?
- Does the explanation provide any valuable insight about the model's reasoning?
- What are the performances of the proposed model?

1.5 Objectives of the Thesis

The first objective of this thesis is to create a deep learning system that can interpret the results to detect lung cancer. These specific objectives include:

- Designing a deep learning model that would be able to classify images of lungs as either normal, benign, or malignant.
- Adopting explainable artificial intelligence (XAI) to visualize and explain how the model arrives at its predictions.
- Assessing the model's performance using measures such as accuracy, precision, recall, and F1 score.

1.6 Thesis Outline

The remainder of the thesis is organized as follows. A overview of prior research on medical imaging and machine learning techniques for lung cancer diagnosis is given in Chapter 2, with a particular emphasis on explainable artificial intelligence and its implications for medical diagnosis. The study methodology, including the dataset, preprocessing procedures, model training, and explainability approach used for this work, is described in Chapter 3. The experimental procedure, the metrics used to assess the results, the analysis of the data and model performance, and the usage of Grad-CAM visualization techniques are all explained in Chapter 4. Finally, Chapter 5 summarizes the overall findings of the research, outlines the main contributions, discusses the limitations of the study, and suggests possible directions for future work.

2. Literature Review

2.1 Literature Review

Deep learning has proven to be a popular technique for detecting lung cancer using medical imaging techniques, especially CT scans [17]. Various researches aim at increasing the accuracy rate and reducing the number of false alarms while automating the process of diagnosing.

Initially, most researches have utilized convolutional neural networks for classifying lung nodules. For example, Joshua et al. [18] suggested using a simple 3D CNN model along with Grad-CAM to classify lung nodules with an accuracy rate of 97.17%. In another study, Saravanaprasad et al. [19] utilized a 3D CNN along with an SVM model as part of their computer-aided diagnosis system, with an accuracy rate of 94%. This shows how deep learning can help detect informative features from medical images. Nevertheless, most of these models are computationally intensive and have been trained using a limited number of data samples.

To further improve performance, several studies introduced hybrid and multi-stage models. Hamidreza et al. [20] combined GAN, reinforcement learning, and deep networks for better segmentation and detection. Wankhade et al. [21] proposed a hybrid model using 3D-CNN and RNN to perform multiple tasks in a single framework. Similarly, Naseer et al. [22] and Naseer et al. [23] used modified CNN models with SVM classifiers, with Naseer et al. [22] reporting 97.98% accuracy. Although these approaches achieve high performance, they increase model complexity and computational cost.

Segmentation and classification have also been studied together. Crasta et al. [24] developed a framework combining 3D-VNet and 3D-ResNet, which achieved 99.2% classification accuracy while maintaining robust performance across different nodule characteristics. Yahia et al. [25] introduced a transformer-based UNETR model

combined with classification, showing strong performance on 3D CT images. These methods improve detection accuracy but require high computational resources and are difficult to deploy in real clinical settings.

Enhancing detection pipelines was the focus of some research. Naseer et al. [22] employed a multi-stage strategy using U-Net for classification after segmentation. A multi-scene deep learning system was presented by Zhang et al. [26] with the goal of enhancing lung nodule identification while reducing false positive predictions. For detection and severity analysis, Asuntha et al. [27] created a hybrid system that combines CNN, feature extraction, and optimization. Nevertheless, these systems rely on several processing stages and are not entirely end-to-end. Additionally, ensemble and multi-modal techniques have been investigated in recent studies. A dual-branch model combining image and mask data was proposed by Emad et al. [28], with an accuracy of 98.04%. For multi-class classification, Saha et al. [29] presented a stacked transfer learning model that combines several CNN architectures. Although these methods perform better, they need larger datasets and more processing power.

Efficiency has also been considered in some studies. Haziq et al. [30] proposed a sequential CNN that reduces computational cost while achieving 95.34% accuracy. However, generalization across datasets remains a challenge.

Abe et al. [31] developed an ensemble CNN model that achieves 98.17% accuracy in lung cancer diagnosis using small CT scan datasets. Their findings suggest that ensemble-based deep learning models can support radiologists in identifying early-stage lung nodules more effectively.

Rai et al. [32] proposed a deep learning-based lung cancer screening framework using the IQ-OTH/NCCD dataset and evaluated the performance of a custom CNN alongside several transfer learning models. By integrating SHAP for interpretability, the study achieved 97.3% accuracy while prioritizing balanced precision and recall for clinical reliability.

Elhassan et al. [33] proposed a dual-model approach combining CNN for classification and YOLOv8 for tumor detection, enhanced with DCGAN-based augmentation. The method achieved 97.67% accuracy and IoU 0.85, but increases complexity and lacks multi-class classification and interpretability.

In addition to model development, several review studies summarized the field. Wang et al., Gao et al., and Tan et al. [34, 35, 36] highlighted that CNN-based

models dominate lung cancer detection research. They also pointed out common challenges such as lack of standardization, limited datasets, and poor generalization. Liu et al. [37] further showed through meta-analysis that AI-based models achieve high diagnostic performance, but variability across studies still exists.

However, there are still some challenges that remain. Most of the studies available mainly focus on increasing accuracy while providing little to no interpretation for their results. This lack of interpretability limits their use in clinical environments where trust and transparency are important. Moreover, many studies focus only on binary classification, while practical diagnosis requires distinguishing between normal, benign, and malignant cases. In addition, many models rely on complex architectures or multi-stage pipelines, making them difficult to deploy. Limited dataset diversity and lack of external validation also affect the generalization of these models.

Therefore, there is a need for a deep learning-based system that can perform accurate multi-class lung cancer detection and provide interpretable outputs to support clinical decision-making.

Table 2.1: Summary of selected recent studies on lung cancer detection

Author (Year)	**Method**	**Dataset**	**Task**	**Result & Strength**	**Limitations**
Crasta (2024)	3D-VNet + 3D-ResNet	LUNA16	Seg+Cls	99.2% acc; robust performance	No XAI, binary focus
Saravanaprasad (2024)	3D CNN + SVM	Kaggle CT	Cls	94% acc; effective CAD	No interpretability, not multi-class
Hamidreza (2024)	GAN + RL + DL	LIDC-IDRI	Seg+Det	IoU 0.97; strong detection	Complex pipeline, no XAI
Eldho (2024)	3D-DLCNN + Mask-RCNN	Clinical CT	Det+Sev	93% acc; low false positives	No interpretability
Saha (2024)	Ensemble CNN	Kaggle CT	Multi-cl	91% acc; stable classification	No XAI, high complexity
Tyagi (2023)	3D Dense CNN	CT datasets	Staging	97% acc; detailed staging	Complex model, no XAI
Wankhade (2023)	3D-CNN + RNN	LIDC-IDRI	Multi-tk	Multi-task capability	No explainability, unclear validation
Yahia (2023)	UNETR (Transformer)	MSD CT	Seg+Cls	98.7% acc; advanced architecture	High complexity, no XAI
Naseer (2023)	U-Net + AlexNet + SVM	LUNA16	Pipeline	97.9% acc; full pipeline	Multi-stage, not interpretable
Haziq (2025)	Sequential CNN	Histopath.	Cls	95% acc; efficient model	No XAI, limited class scope
Emad (2026)	CNN + DNN (Dual-branch)	LIDC-IDRI	Multi-mo	98% acc; strong fusion	Complex, no interpretability

3. Methodology

3.1 Introduction

The aim of this research is to develop a deep learning model that can precisely forecast lung cancer using CT scans [38, 39]. Deep learning techniques have become increasingly popular in medical image analysis due to their ability to extract intricate features and deliver high accuracy [5, 40]. Diagnosing lung cancer in CT images presents various challenges, such as the variability in cancer size, shape, density, and location [41, 42]. Furthermore, manually diagnosing cancers can be quite challenging and time-consuming. Deep learning, particularly convolutional neural networks (CNNs), has demonstrated significant effectiveness in cancer diagnosis from medical images [34]. It's unfortunate that the lack of transparency and explainability in deep learning models may render these methods impractical for clinical use. To address this limitation, XAI techniques are increasingly being integrated with deep learning models in the medical decision-making process [43, 44].

Our proposed system will consist of multiple stages of data processing, including preparation, pre-processing, model training, prediction, and explanation. Initially, images from CT scans will be collected and categorized into specific groups. Following this, pre-processing steps will be undertaken to improve the quality of the images. A deep learning model will then be trained on the dataset to tackle the classification task. Finally, explainability will be attained through the use of XAI techniques.

The primary objectives of this methodology consist of the following:

- To develop an automated system for detecting lung cancer through CT image scans.

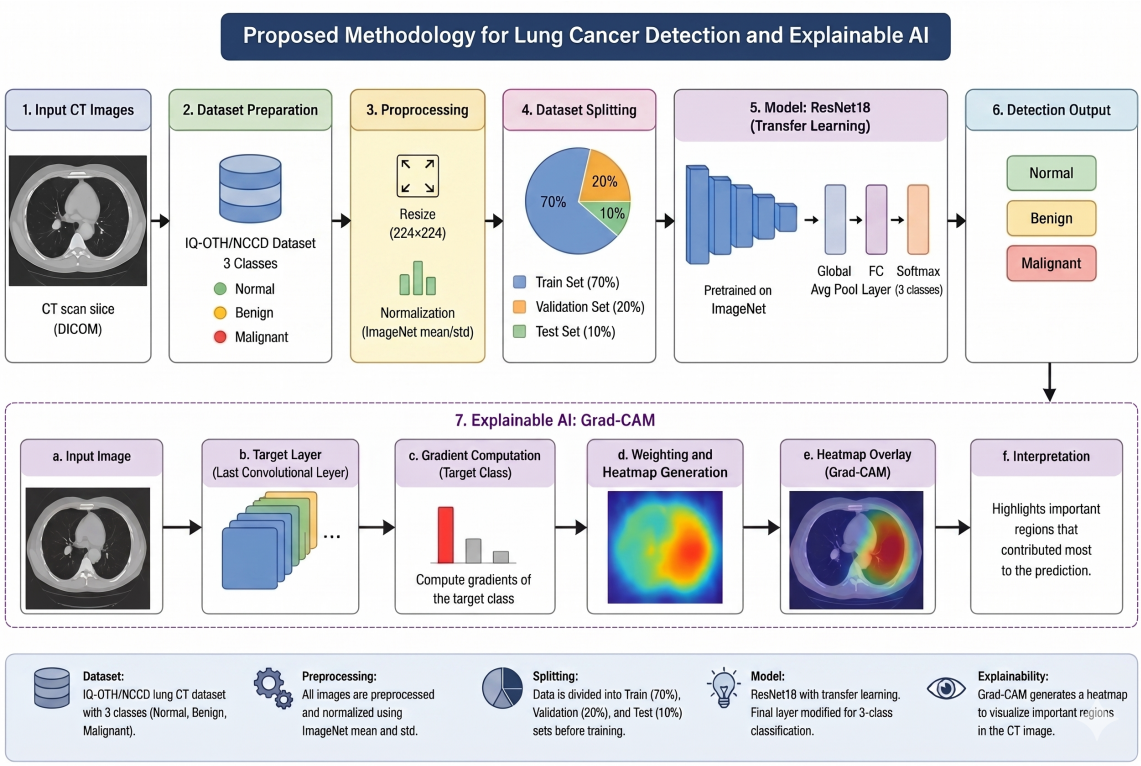


Figure 3.1: Proposed methodology of lung cancer detection system with explainable AI.

- To classify lung images into three categories: normal, benign, and malignant.
- To enhance the precision of models by utilizing effective preprocessing techniques.
- To apply explainable artificial intelligence for interpreting the model's outputs.

3.2 Dataset

For our study, we used a publicly available dataset called IQ-OTH/NCCD [45] to train, validate, and test our lung cancer detection model. It was collected from the Iraq-Oncology Teaching Hospital and the National Center for Cancer Diseases [46] and contains CT scan images from both healthy people and those diagnosed with lung cancer. Doctors often use CT scans to detect lung cancer because these scans give clear, detailed views of the lungs. This dataset has a total of 1,190 CT scans, which fall into three groups:

- Normal - 55 cases
- Benign - 15 cases
- Malignant - 40 cases

Each case includes many CT slices, anywhere from 80 to 200 images, showing different cross-sections of the lung. The original images were saved in DICOM format but were later converted into regular image files to make them easier to work with [47].



(a) (b) (c)

Figure 3.2: Some samples of CT images from the dataset where (a) normal, (b) benign, and (c) malignant cases.

However, the dataset is small and has imbalanced classes. This can make it hard for the model to learn properly.

3.3 Data Pre-processing

Reprocessing is an essential phase in the handling of medical images [48], making sure that the input data is appropriate for the deep learning model's application. The success of machine learning algorithms, particularly in medical image analysis, is largely dependent on the efficacy of preprocessing [49].

3.3.1 Image Resizing

Images are standardized in size for all the input pictures in order to maintain uniformity in the entire database. This process is necessary since the model used

requires consistent size inputs, besides making it easier for computation and hence efficient during training [50, 49]s

Let the original image size be $W_{orig} \times H_{orig}$. The image is resized to a target dimension $W_{target} \times H_{target}$.

3.3.2 Normalization

Normalization is used to adjust pixel values so they fall between 0 and 1. This is done by simply dividing each pixel value by 255.

$$I_{scaled} = \frac{I_{pixel}}{255} \tag{3.1}$$

This process helps in stabilizing the training process and improving convergence of the model [51].

3.3.3 Data Augmentation

To tackle the problems faced with having a small training set and class imbalance, methods such as data augmentation were implemented. Data augmentation increases the size of the training set artificially and may help reduce overfitting [37]. The following data augmentation methods were used:

The following augmentation techniques are used:

- Rotation: The image undergoes rotation at a randomly selected angle in a certain range, which ensures that the model is invariant to the orientation of the medical feature.
- The image is flipped horizontally, essentially increasing the training set by a factor of two and preventing the model from being oriented towards a particular direction for diagnosis.
- Zooming: Random zoom in and out functions are performed on the images, which enables the network to learn features at varying distances from the camera to subject.
- Shift: Random horizontal and vertical shifting is performed, which trains the model to learn about the location of the target region regardless of its placement in the image.

3.3.4 Dataset Splitting

After preprocessing, the dataset is divided into training, validation, and testing sets. In this study, we split the data 70% for training, 20% for validation, and 10% for testing, before applying data augmentation. After splitting, we applied an augmentation technique to train set only to ensure that augmented images derived from a single original image are not distributed across both training and validation sets. Such leakage could lead to overly optimistic performance estimates, as the model may indirectly learn from similar samples present in both sets. The testing set remains unchanged to provide a fair and unbiased evaluation of the model's performance on unseen data.

3.4 Model Selection

Choosing the right model is essential for obtaining dependable results in medical image classification. We trained and evaluated several models alongside our proposed model, including Xception, MobileNetV2, CoAtNet, and an Ensemble method (Max Voting). In our thesis work, we utilized a transfer learning approach founded on the ResNet18 architecture.

ResNet (Residual Network) is a sophisticated deep convolutional neural network [52] that incorporates residual connections to tackle the vanishing gradient issue [53], facilitating the training of deeper networks effectively. Among the various versions, ResNet18 has been selected for its optimal compromise between performance and computational effectiveness [54].

In contrast to developing a model from the ground up, transfer learning capitalizes on the knowledge [55] gained from models that have been pre-trained on extensive datasets like ImageNet [56]. This approach is especially advantageous in medical imaging applications where available dataset sizes are often limited. By utilizing weights that have been pre-trained, the model can learn essential low-level features such as edges and textures, which are pertinent in CT images.

Alternative models like Vision Transformers (ViT) [57] demand much larger datasets for effective generalization and are more resource-intensive. As a result, ResNet18 is preferred for this research due to its efficiency and strong efficacy with smaller datasets.

In this experiment, the final layer of the pre-trained ResNet18 model is modified to produce outputs for three classes: normal, benign, and malignant. The remaining layers maintain their original weights, enabling the model to effectively learn to classify lung cancer.

3.5 Proposed Model Architecture

The proposed framework is built upon the ResNet18 architecture [52]. It is composed of several convolutional layers featuring residual connections, succeeded by a fully connected layer for the purpose of classification [52].

3.5.1 Residual Learning

The central concept of ResNet revolves around residual learning. Rather than directly predicting the output, the network focuses on learning the residual or the difference between the input and the output. This concept is illustrated in Eq. 3.2.

$$y = F(x) + x \tag{3.2}$$

In this equation, x is the input, $F(x)$ is the residual function, and y is the final output. This helps in preserving gradient flow and improving training stability [52].

3.5.2 Network Modification

The pre-trained ResNet18 model is modified by replacing its final fully connected layer, which is defined by Eq.3.3.

$$y = Wx + b \tag{3.3}$$

Here, W and b are parameters that the model learns during training. The output y has three values, one for each class: normal, benign, and malignant.

3.5.3 Activation and Output

The output of the last layer undergoes a transformation using the softmax function. It is defined by Eq.3.4.

$$P(y = i) = \frac{e^{z_i}}{\sum_j e^{z_j}} \tag{3.4}$$

This enables multi-class classification [58].

3.6 Model Training and Evaluation

The model is trained using the preprocessed dataset with three classes. During training, the model performs three main steps: forward propagation, calculating the loss, and backpropagation.

3.6.1 Loss Function

We use categorical cross-entropy as the loss function. This is shown in Eq. 3.5.

$$L = -\sum_i y_i \log(\hat{y}_i) \tag{3.5}$$

3.6.2 Optimizer

For training, we use the AdamW optimizer. It combines an adaptive learning rate with weight decay regularization to improve performance [59]. It is defined by Eq.3.6.

$$\theta = \theta - \eta \cdot \hat{m}/(\sqrt{\hat{v}} + \epsilon) \tag{3.6}$$

where θ represents model parameters.

3.6.3 Learning Rate Scheduling

We also use a ReduceLROnPlateau scheduler. It lowers the learning rate when the validation loss stops getting better. This helps the model converge more smoothly.

3.6.4 Training Setup

The model is trained using mini-batch gradient descent with the following configuration:

- Batch size: 32

- Learning rate: 1×10^{-4}
- Epochs: 30
- Input size: 224×224

The model is trained using a GPU when one is available to speed up the computation process. Following the training, we assess the performance of the proposed model with a distinct test dataset that was not utilized during the training or validation phases. Our evaluation employs various metrics, including Accuracy, Precision, Recall, and AUC-ROC [60, 61]. Additionally, we utilize a confusion matrix to analyze the model's performance across each class. Together, these metrics provide a comprehensive view of the model's effectiveness in detecting lung cancer across three categories.

3.7 Explainable AI using Grad-CAM

While deep learning models may demonstrate high accuracy when detecting abnormalities based on medical images, they often remain black-boxes. The inability to determine why the model makes particular predictions is a problem. For clinical settings, interpretability is especially important because it helps build confidence in a model and assists medical experts in making well-informed decisions. To ensure our model is interpretable, we use Gradient-weighted Class Activation Mapping (Grad-CAM).

Grad-CAM is one of the most popular methods in explainable AI [8]. It determines the area of the image that significantly influence the prediction of the machine learning model. The method allows us to visualize such regions by creating heatmaps of input images.

To use Grad-CAM, we feed the trained ResNet18 model with the final convolutional layer and generate a heatmap for each CT image input. We then combine this heatmap with the original image, which helps to see what parts of the lungs are particularly attention-grabbing for the model. For instance, in malignant cases, the model usually pays special attention to the suspected nodules or other problematic parts of the image.

The main advantages of using Grad-CAM are as follows:

- Helps to visualize the regions influencing the model decision [62].
- Assists in ensuring that the machine learning model focuses on clinically relevant parts of the image.
- Helps medical experts in interpreting the model prediction [63].
- Makes the system more transparent and reliable [64].

By integrating Grad-CAM, the proposed system not only performs accurate detection of lung cancer types but also provides meaningful explanations, addressing one of the key limitations of deep learning models in healthcare.

3.8 Financial Planning for the Project

Table 3.1 outlines the estimated budget for the Reliable Lung Cancer detection from Medical Images and Model Interpretability project. This includes key components like personnel, hardware, software, and data acquisition to ensure successful project completion.

3.9 Gantt Chart of our Thesis work

Figure 3.3 shows the project schedule over 10 months. It includes six different steps and how long each step is expected to take. It begins with the Literature Review and Research Gap Identification from August to May. Preliminary Experiments are conducted in August-January to assess feasibility. Dataset collection takes place in November and December to gather the data needed for the next stage. Model development happens from January to March. Finally, thesis writing and publication run from November through May, making sure all findings are properly documented.

3.10 Conclusion

This chapter discusses the detection process used for lung cancer in this study. The use of the IQ-OTH/NCCD dataset with CT scan images classified as normal,

Table 3.1: Budget Breakdown for the lung cancer detection with explainable AI.

Budget Type	Description	Unit	Total (BDT)
Hardware	Purchase of storage devices (SSD, external hard drives) for seamless data storing.	02	20,000
Software	Premium membership of Lucidchart and Overleaf for designing methodology, architecture, and writing book.	02	10,000
Data Acquisition	Expenses for acquiring datasets (lung images), annotating them, and applying augmentation for training.	01	30,000
Cloud Computing	Usage of cloud services (such as Colab Premium, Kaggle Premium) for model training and storage of large datasets.	02	20,000
Miscellaneous	Miscellaneous costs including transport, printing, reproduction costs, mobile bills, and small research materials.	01	10,000
Grand Total			**90,000**

benign, and malignant is done. Preprocessing was performed to ready the data for the learning process through a series of steps such as resizing and normalization.

The transfer learning algorithm using ResNet18 as a basis was used for classifying the CT scan images into three different categories. With limited medical information available, pre-trained weights were used. Optimized learning algorithms and suitable evaluation metrics were used to train and get good results from the model.

Noting that deep learning models can be very difficult to interpret, Grad-CAM was added as a way of making the model more understandable. It helps by highlighting parts of the CT scan image that are contributing to the model decision-making process.

Overall, the method proposed is very efficient in detecting lung cancer.

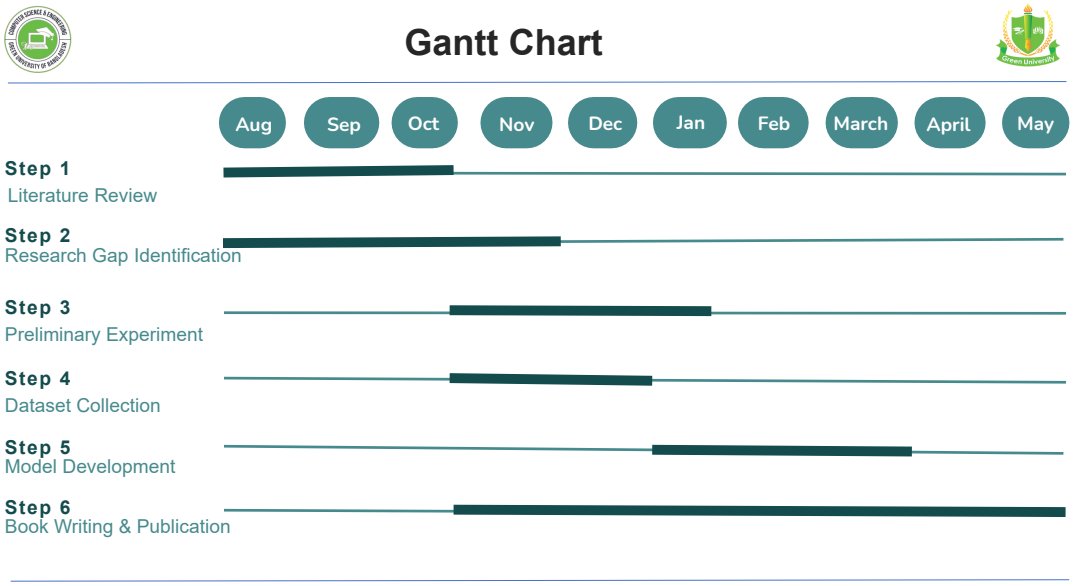


Figure 3.3: Gantt chart of our system development.

4. Result Analysis and Discussion

4.1 Introduction

This section outlines the outcomes and analysis of our lung cancer detection system. We evaluated the effectiveness of the model using standard classification metrics on the IQ-OTH/NCCD dataset. The results are examined from various perspectives, including the model's training effectiveness, the accuracy of its classifications, and the ease of interpretation.

The purpose of this section is to assess how proficiently our ResNet18-based model can categorize CT images into normal, benign, and malignant categories. Additionally, we assess the model's interpretability through Grad-CAM, which aids in understanding which areas of an image impacted the model's predictions.

4.2 Experimental Setup

We developed our system leveraging PyTorch, a deep learning framework [65]. The experiments utilized the IQ-OTH/NCCD lung cancer dataset, which comprises CT scans categorized into three classes: normal, benign, and malignant. Prior to the training phase, the images were adjusted to a size of 224×224, and normalized according to the mean and standard deviation derived from ImageNet data. We divided the dataset into 70% for training, 20% for validation, and 10% for testing. This division was carried out before applying any data augmentation to prevent data leakage.

In this research, we employed a ResNet18 model with transfer learning. We initialized with weights that had been pre-trained on ImageNet. Then, we modified the final fully connected layer so it can output three classes.

The model was trained using the settings below:

- Optimizer: AdamW
- Learning rate: 1×10^{-4}
- Batch size: 16
- Number of epochs: 30
- Loss function: Cross-Entropy Loss

The ReduceLROnPlateau schedule is used for adapting the learning rate according to the validation loss. If available, the experiment will be done using GPU acceleration.

4.3 Performance Evaluation Metrics

The following list of metrics was used for measuring the performance of the developed model. This helps determine the effectiveness of the proposed model, especially in image classification into three categories.

4.3.1 Accuracy

Accuracy reflects the overall correctness of the model, calculated using Eq.4.1

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{4.1}$$

4.3.2 Precision

Precision represents the portion of positive predictions made by the model that are truly positive, as quantified by Eq.4.2.

$$Precision = \frac{TP}{TP + FP} \tag{4.2}$$

4.3.3 Recall

Recall (also known as Sensitivity) represents the percentage of true positives captured by the prediction. Its calculation formula is provided in Eq.4.3.

$$Recall = \frac{TP}{TP + FN} \tag{4.3}$$

4.3.4 F1-Score

The F1-Score combines the measurements of precision and recall through the use of a harmonic mean of both values as expressed in 4.4.

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \tag{4.4}$$

4.3.5 Confusion Matrix

Confusion matrices assess the model's performance on each class separately by analyzing the number of correct and incorrect predictions.

4.3.6 Receiver Operating Characteristic (ROC) Curve

This approach evaluates the model's discriminative power by plotting the ROC curve for each class using the one-versus-rest method. The Area Under the Curve (AUC) indicates the model's ability to discriminate [66].

4.3.7 Explainability using Grad-CAM

Grad-CAM is used to highlight specific regions of the CT image that influence the model's predictions, allowing for verification of whether it concentrates on the pertinent areas of lungs.

4.4 Training Performance

The curves representing the training and validation loss are shown in Figure 4.1. As shown in the graph, both the lines show a consistent decline with respect to epochs, indicating that the model learns continuously without undergoing overfitting. Therefore, it can be inferred that the selected learning rate and parameters of the

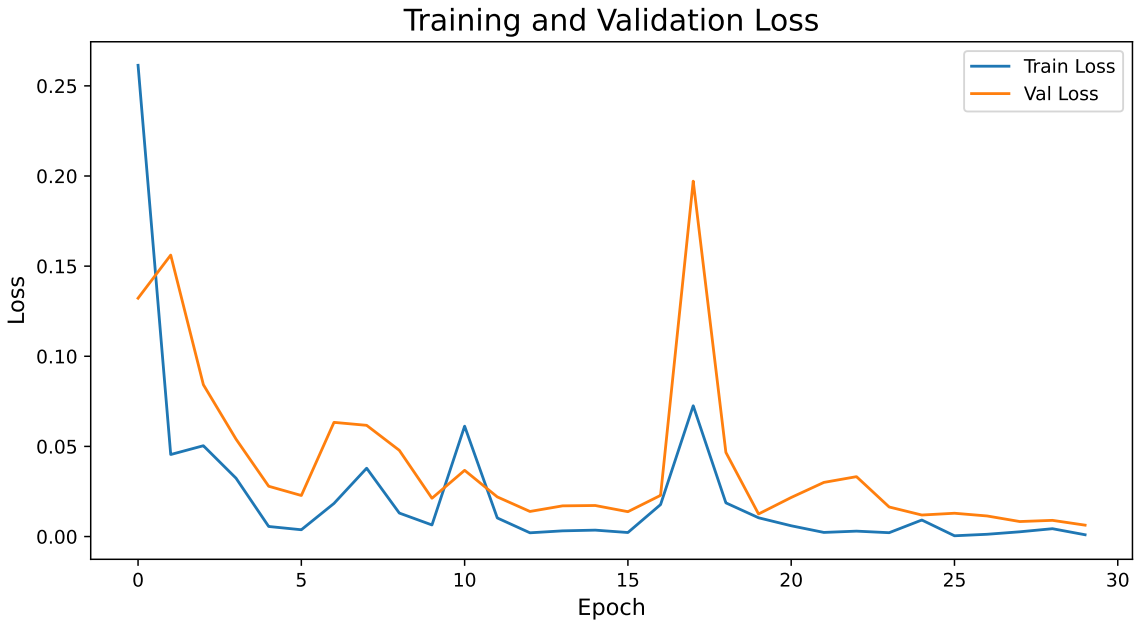


Figure 4.1: Training and validation loss curves over epochs showing stable convergence of the proposed model

optimizer are suitable for the task at hand. Likewise, Figure 4.2 represents the training and validation accuracy curves. The validation accuracy stays close to the training accuracy, which suggests that the model generalizes well to new data [67].

4.5 Classification Results

Confusion matrix has been shown in Figure 4.3. The model perfectly classified all malignant and normal images while just one benign image was wrongly classified. This shows that the model performed remarkably well, especially when considering vital categories such as malignant cancer. Correct identification of malignant cancer is very important as far as early diagnosis is concerned [2].

4.5.1 Classification Report

As seen in Figure 4.4, the classification report gives a detailed evaluation of the performance of the model. Overall, the accuracy rate of the model is 99%. However, in terms of recall of benign, a lower score shows that some of the benign cancers were wrongly classified because of class imbalances in the data set.

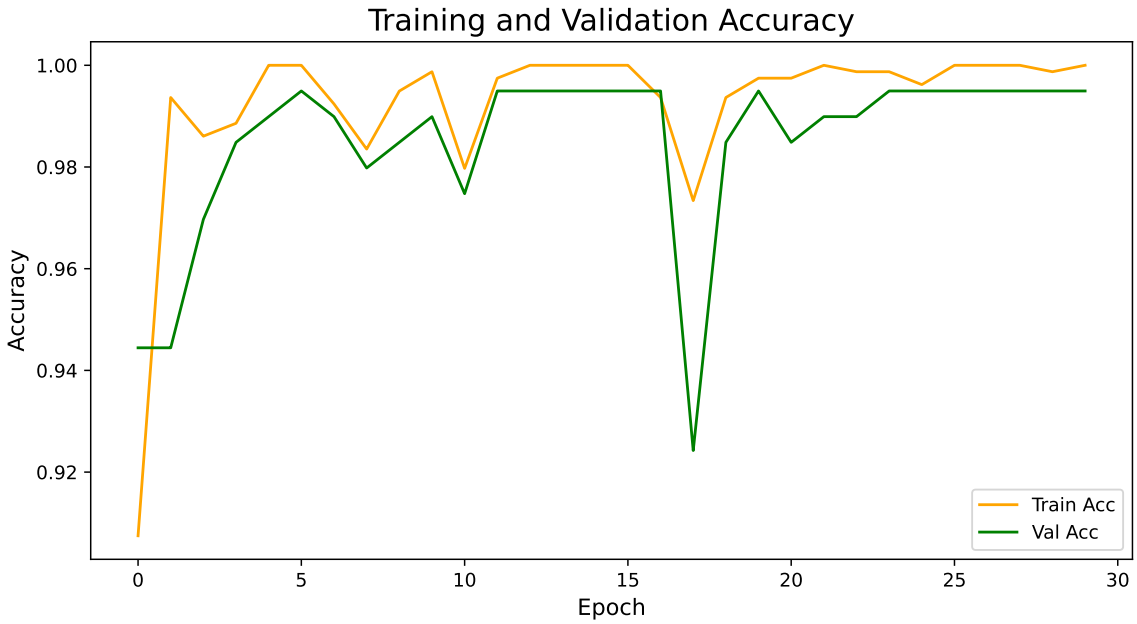


Figure 4.2: Training and validation accuracy curves demonstrating consistent performance and good generalization

4.6 ROC Curve Analysis

The ROC curves for all three classes are shown in Figure 4.5.

The model achieves excellent AUC scores:

- Malignant: AUC = 1.000
- Normal: AUC = 0.997
- Benign: AUC = 0.993

These results indicate that the model has strong discriminative capability across all classes.

4.7 Explainability Analysis with Grad-CAM

The visualization outcome using Grad-CAM is presented in Figure 4.6 below. The heat maps emphasize parts of the CT images that have the most significant impact on the decision-making of the model. For a malignancy situation, the model focuses on the unusual lungs that are relevant to the diagnosis process. It shows that the

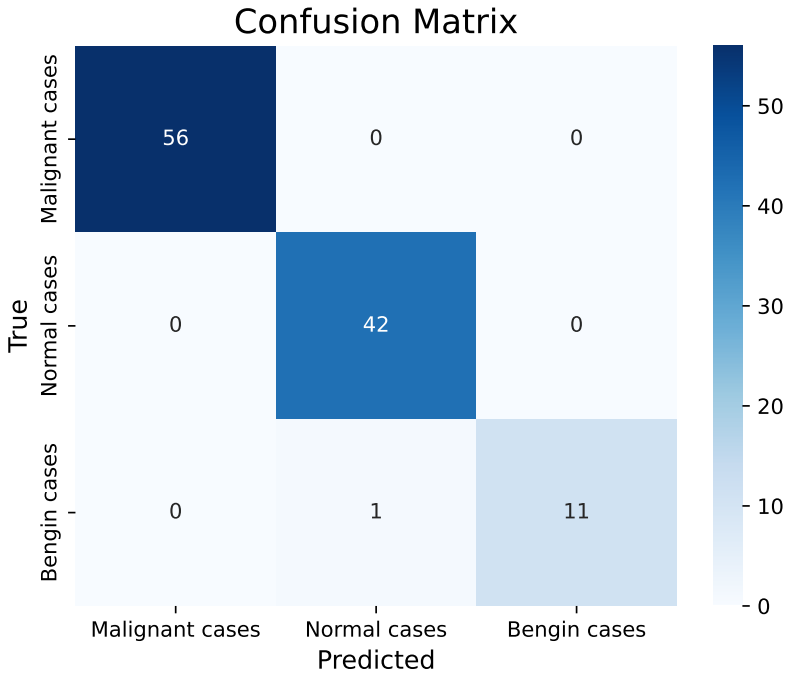


Figure 4.3: Confusion matrix showing how well the model classifies normal, benign, and malignant cases

model is accurate and explainable since it makes decisions based on anatomically relevant parts of the CT images [32].

4.8 Performance Comparison with Other Models

To evaluate the effectiveness of the proposed approach, several deep learning models were experimented with on the same dataset. The classification accuracies obtained from different architectures are presented in Table 4.1.

Table 4.1: Performance comparison of different deep learning models

Model	**Accuracy**
Xception	85.00%
MobileNetV2	82.50%
CoAtNet	98.89%
Ensemble (Max Voting)	93.89%
ResNet50	95.56%
Proposed ResNet18 + Grad-CAM	**99.00%**

Classification Report:

	precision	recall	f1-score	support
Malignant cases	1.00	1.00	1.00	56
Normal cases	0.98	1.00	0.99	42
Bengin cases	1.00	0.92	0.96	12
accuracy			0.99	110
macro avg	0.99	0.97	0.98	110
weighted avg	0.99	0.99	0.99	110

Figure 4.4: Performance metrics of the proposed model including precision, recall, and F1-score for each class

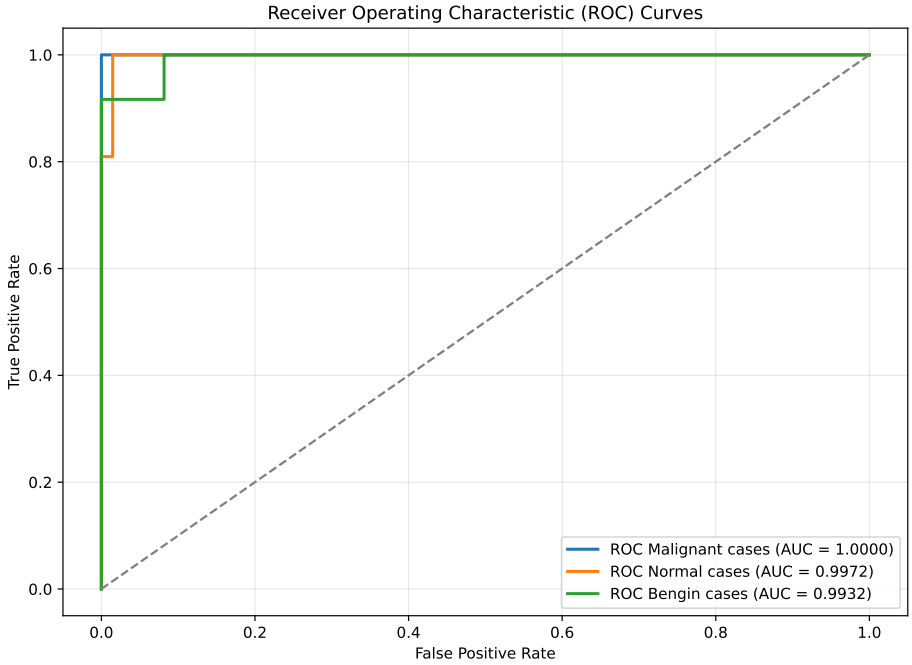


Figure 4.5: ROC curves for multi-class classification showing high AUC values for all classes

From Table 4.1, it can be observed that the proposed ResNet18-based model achieves the highest classification accuracy among the evaluated models. MobileNetV2 and Xception show comparatively lower performance, while the ensemble and ResNet+ models improve the results significantly. CoAtNet achieves competitive performance with 98.89% accuracy; however, the proposed model slightly outperforms it while also providing explainability through Grad-CAM

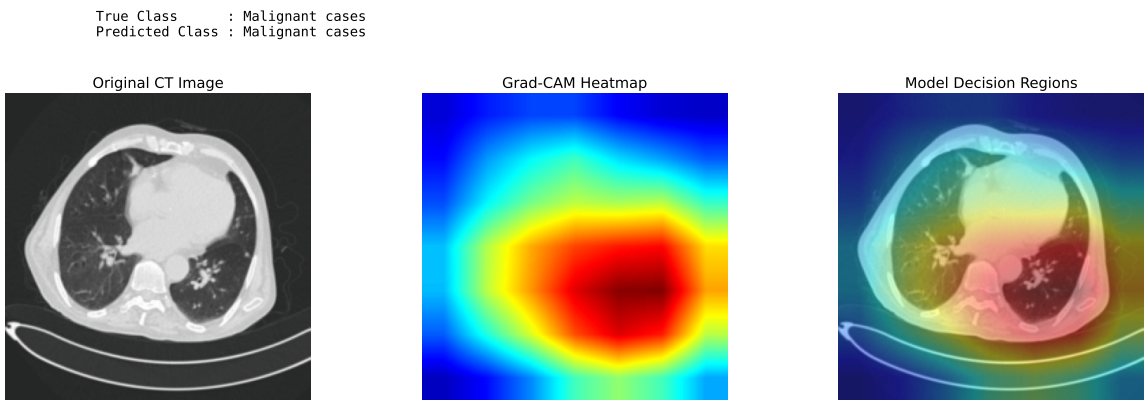


Figure 4.6: Grad-CAM visualization highlighting important regions influencing the model's prediction in CT images

visualization.

4.9 Comparative analysis with existing works

Table 4.2: Comparison with existing works with proposed model

Study	Method	Dataset	XAI	Accuracy
Saravanaprasad (2024)	3D CNN + SVM	Kaggle CT	No	94%
Abe (2025)	Ensemble CNN	IQ-OTH/NCCD	No	98.17%
Haziq (2025)	Sequential CNN	Histopathology	No	95.34%
Emad (2026)	CNN + DNN	LIDC-IDRI	No	98.04%
Rai (2026)	ResNet152 + SHAP	IQ-OTH/NCCD	Yes	97.3%
Proposed	**ResNet18 + Grad-CAM**	**IQ-OTH/NCCD**	**Yes**	**99.00%**

Table 4.2 presents a comparison between the proposed model and recent studies. Current methods can be categorized into three distinct groups.

Firstly, hybrid and complex models such as those developed by Saravanaprasad et al.[19] and Emad et al.[28] achieve accuracy levels ranging from moderate to high (94%–98.04%), but they depend on multi-stage or dual-branch architectures, which lack explainability. Similarly, Abe [31] attains a high accuracy of 98.17% by employing an ensemble CNN, though this results in increased computational complexity and diminished interpretability.

Second, efficient or alternative models like Haziq et al.[30] improve computational efficiency but are evaluated on different data modalities and do not include explainability, limiting their applicability to CT-based diagnosis.

Third, explainable models such as Rai et al.[32] incorporate XAI using SHAP, but use deeper architectures (ResNet152) with higher computational cost and lower accuracy (97.3%).

On the other hand, our ResNet18-based model reaches the highest accuracy at 99.00%. It keeps a simple design and includes Grad-CAM to make its predictions easy to see and understand. This gives a good mix of strong performance, low complexity, and clear explainability, which makes it more useful for real-world clinical settings.

4.10 Discussion

According to the experimental findings, the suggested transfer learning model utilizing the ResNet18 architecture demonstrates effectiveness in identifying multi-class lung cancer from CT scan images. The model achieves an accuracy score of 99%, indicating its ability to leverage discriminative features for categorizing images as normal, benign, or malignant. Through the implementation of transfer learning, the model is able to take advantage of pre-trained feature representations, enhancing learning efficiency even when dealing with a limited dataset of medical imaging [68].

The patterns observed in training and validation indicate a steady decline in loss during the course of the experiment, achieving stable convergence after 8 training epochs. This suggests a lack of overfitting, and confirms that the model effectively generalizes the knowledge gained from the training data. In addition, by incorporating AdamW optimizer and learning rate scheduling, we managed to maintain consistency in training and avoid possible overtraining.

Additional insights into the behavior of our model were obtained through analysis of confusion matrix and classification report. It should be noted that the system correctly classified all malignant and normal cases from the test data. This is especially important since correct detection of malignant cases in the early stages is crucial for diagnosis and planning of treatment [2]. The lowest score in terms of recall was observed for the benign class due to one incorrectly classified case. This could happen due to visual similarities between benign and malignant nodules and difficulties in recognizing differences in CT scans. Moreover, benign tumors make up fewer examples in the dataset compared to malignant and normal tumors.

Finally, ROC curve analysis confirms that our proposed solution performs well, as all classes achieved very high AUC scores, with the malignant class reaching 1.000, which speaks of excellent separation capacity of the model.

Analysis of Grad-CAM heatmaps revealed that the model is focused only on important parts of the lung image. For example, in malignant tumors, the model identifies parts of the body, where malignant changes occur and can be seen in the CT scans.

When comparing our solution to recently published papers, we should note that our model proved to be comparable in terms of performance while having a much simpler architecture. Existing models require either complex ensembles or hybrid systems to achieve similar results, whereas our lightweight architecture combined with Grad-CAM explainability provides competitive classification results.

4.11 Conclusion

The performance of the lung cancer classification model is assessed through various parameters within this chapter. The model shows excellent accuracy and sufficient capability of correctly classifying all three classes. The integration of Grad-CAM enhances interpretability, addressing a key limitation of deep learning models. The results indicate that the proposed approach is both accurate and reliable, making it a promising solution for automated lung cancer detection.

5. Conclusion

5.1 Concluding Remarks

The study being proposed presents a deep learning-driven system designed to categorize CT scans based on the presence of lung cancer. The transfer learning approach is utilized within a pre-trained ResNet18 [52] network to sort the images into one of three categories: normal, benign, and malignant [69].

The results show that the model performed really well, hitting an average accuracy of 99%. Additionally, the model demonstrates remarkable precision in identifying cancer in patients, which is crucial for their treatment and care. Furthermore, Grad-CAM has been incorporated into the system to help users comprehend the reasons behind certain segments of the CT scan impacting the model's predictions.

In summary, our proposed way gives an effective and easy-to-interpret solution for automatic lung cancer detection from CT images.

5.2 Research Contributions

- A system for detecting lung cancer using deep learning technology was built by applying ResNet18 with transfer learning to execute multi-class classification.
- The IQ-OTH/NCCD dataset was employed [70], along with some preprocessing techniques, such as resizing and normalization.
- This approach demonstrated exceptional efficacy in detection, attaining an accuracy of 99
- The concept of Grad-CAM was implemented to provide visual explanations, overcoming the issue of lack of interpretability encountered in most models.

- Comparison with modern researches was also done, revealing that this model maintains a good trade-off between accuracy, efficiency, and explainability.

5.3 Limitations of the Study

Despite the success that the experiment yields, there are several limitations associated with it:

- Small size of the dataset might hinder generalization by the model.
- There is no evaluation on another dataset outside the one provided for the assessment of the model's performance.
- There are some minor errors made during classification of benign lesions due to imbalanced nature of the classes involved.
- The analysis is purely focused on the use of images without any input from the medical record.

5.4 Future Work

- Using a bigger and more varied dataset to improve the robustness of the algorithm.
- Utilizing MRI images obtained from the clinical environment and developing an algorithm to compare its efficiency with that of CT images.
- Using multimodal data that would include both clinical and demographic data.
- Developing a decision support algorithm that can be used in the clinical setting.

5.5 Summary

To conclude, the current thesis shows that the integration of a deep learning model using transfer learning techniques, along with explainable AI, can achieve accuracy along with increased interpretability for the diagnosis of lung cancer. The suggested technique may help doctors make quick and accurate diagnoses.

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